

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
Student Name	Naveenkumar S ¹
Register Number	3122247001038
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

To implement, evaluate, and compare single non-parametric (Decision Tree) and ensemble tree-based (Random Forest) classification algorithms on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset for binary tumor diagnosis, perform feature correlation analysis, execute hyperparameter optimization via 5-fold cross-validation (`GridSearchCV`), and analyze classification metrics, decision rules, impurity measures, and generalization capabilities.

2 Problem Statement

Automated diagnosis of breast cancer from fine needle aspirate (FNA) cell nucleus measurements is a fundamental binary pattern recognition challenge. Given 30 continuous numerical attributes extracted from cell nucleus images, the objective is to construct predictive models that accurately classify tumor instances into legitimate benign lesions (Benign = 0) or malignant tumors (Malignant = 1).

Input: Feature vector $\mathbf{x} \in \mathbb{R}^{30}$ representing cell nucleus characteristics (mean, standard error, and worst values for radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension).

Output: A binary class label $y \in \{0, 1\}$ indicating Benign (0) or Malignant (1).

Objective: Maximize classification Accuracy, Precision, Recall, and F1-score while evaluating the impact of impurity criteria (`gini`, `entropy`), tree depth (`max_depth`), leaf sample split bounds (`min_samples_split`, `min_samples_leaf`), ensemble size (`n_estimators`), and feature subsampling (`max_features`).

¹GitHub Repository: <https://github.com/naveensugi14/MACHINE-LEARNING>

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic))
Number of Samples	569 instances
Number of Features	30 numerical input features + 1 binary target class attribute
Feature Breakdown	10 Mean metrics, 10 Standard Error (SE) metrics, 10 Worst metrics
Class Distribution	Benign (B): 357 (62.7%), Malignant (M): 212 (37.3%)
Missing Values	0 missing entries across all features
Train-Test Split	80% / 20% (455 training samples / 114 test samples, <code>random_state = 42</code>)
Feature Scaling	Tree-based models are scale-invariant; raw feature scaling preserved

4 Exploratory Data Analysis

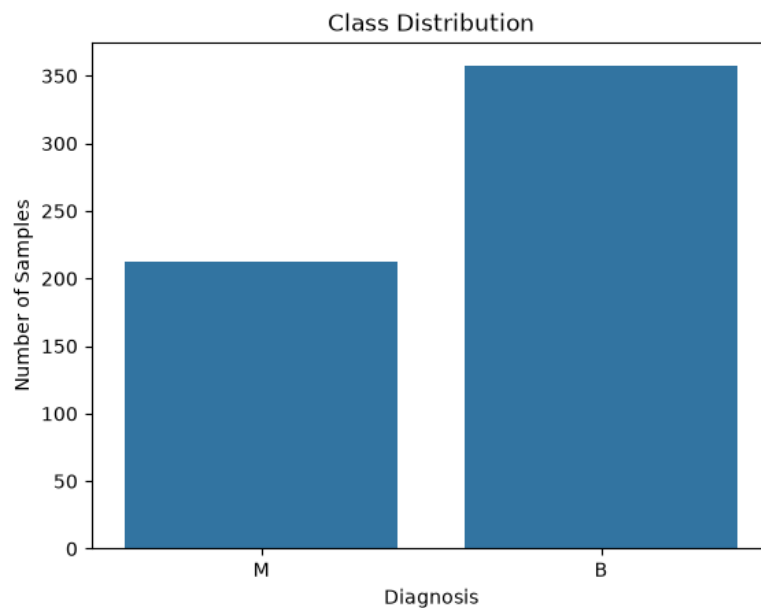


Figure 1: Class Distribution of Malignant (M) vs. Benign (B) samples extracted directly from Ex-5.ipynb.

Inference: The target variable exhibits a well-balanced class distribution with 357 Benign (B) samples (62.7%) and 212 Malignant (M) samples (37.3%). This balanced distribution confirms that overall accuracy, F1-score, and ROC-AUC are appropriate evaluation metrics.

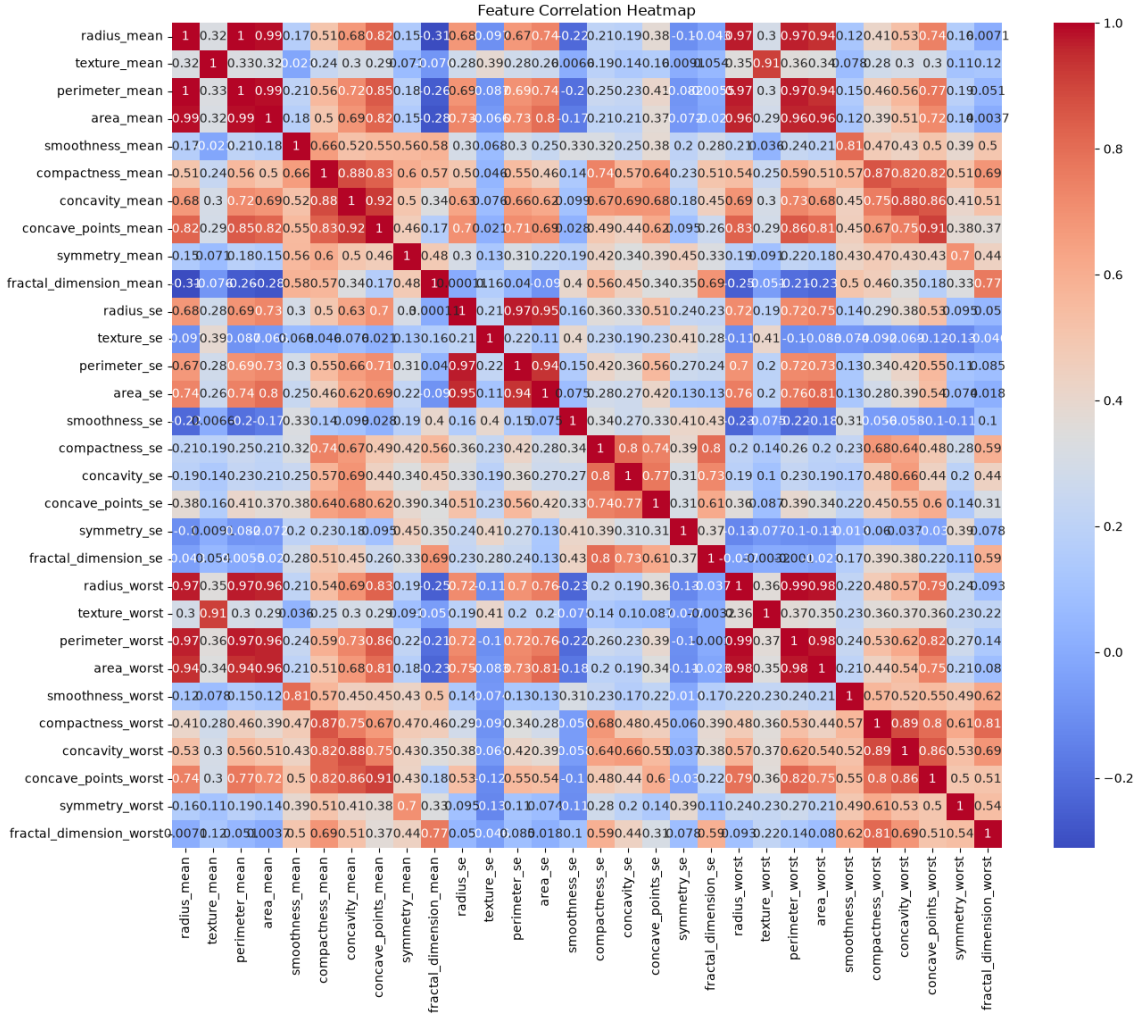


Figure 2: Feature Correlation Heatmap across all 30 numerical attributes extracted directly from `Ex-5.ipynb`.

Inference: Strong positive correlations ($\rho > 0.90$) are observed among geometric attributes such as radius, perimeter, and area. Higher absolute correlation values highlight key predictive signals that drive internal split node selections in decision trees and ensemble models.

5 Data Preprocessing

- **Missing Value Handling:** Verified 0 missing entries across all 30 numerical feature columns.
- **Feature Clean-Up:** Dropped the non-predictive identification column (`id`).
- **Target Encoding:** Applied `LabelEncoder` to convert string diagnosis labels to binary numeric values (Benign (B) $\rightarrow$ 0, Malignant (M) $\rightarrow$ 1).
- **Train-Test Split:** Partitioned the dataset into 80% training (455 samples) and 20% testing (114 samples) sets using stratified sampling (`random_state = 42`).

- **Feature Scaling Invariance:** Decision Trees and Random Forests split feature space orthogonally based on threshold rules, making feature scaling non-essential.

6 Machine Learning Algorithms

1. Decision Tree Classifier

Working Principle: A Decision Tree recursively partitions the input space into axis-aligned hyperplanes based on feature threshold splits to minimize node impurity.

Mathematical Formulation: Impurity at node m with class probabilities p_{mk} is measured using **Gini Index** $G(m)$ or **Entropy** $H(m)$:

$$G(m) = 1 - \sum_{k=0}^1 p_{mk}^2 \quad (1)$$

$$H(m) = - \sum_{k=0}^1 p_{mk} \log_2(p_{mk}) \quad (2)$$

Information Gain: Splits are chosen to maximize impurity reduction:

$$\Delta I = I(m) - \left(\frac{N_{\text{left}}}{N_m} I(m_{\text{left}}) + \frac{N_{\text{right}}}{N_m} I(m_{\text{right}}) \right) \quad (3)$$

Advantages: High interpretability, non-parametric, requires no feature scaling, handles non-linear feature interactions naturally.

Limitations: Prone to high variance and overfitting when grown to unconstrained depth.

2. Random Forest Classifier

Working Principle: Random Forest is an ensemble learning method that builds a forest of B decorrelated decision trees using **Bootstrap Aggregation (Bagging)** and **Random Subspace Feature Selection**.

Ensemble Voting: Each individual tree T_b is trained on an independent bootstrap sample. Split nodes consider a random subset of features $m = \sqrt{p}$ or $m = \log_2 p$. The final class prediction is determined by majority vote:

$$\hat{y}_{\text{RF}} = \text{mode} \left\{ \hat{C}_1(\mathbf{x}), \hat{C}_2(\mathbf{x}), \dots, \hat{C}_B(\mathbf{x}) \right\} \quad (4)$$

Advantages: Reduces single-tree variance, robust against noise, improves generalization performance.

Limitations: Slightly higher computational complexity compared to a single decision tree.

7 Experimental Setup

Python Version	3.11.x
Scikit-Learn Version	1.4.x
Libraries Used	pandas, numpy, matplotlib, seaborn, scikit-learn
IDE	Jupyter Notebook
Operating System	Windows 11
Validation Strategy	5-Fold Stratified Cross-Validation (GridSearchCV)

8 Hyperparameter Selection

Table 1: Hyperparameter Search Space & Optimization Summary

Model	Search Method	Best Parameters
Decision Tree	GridSearchCV	criterion: 'gini', max_depth: 7, min_samples_leaf: 1, min_s
Random Forest	GridSearchCV	bootstrap: False, max_depth: 5, max_features: 'log2', n_est

9 Results

Table 2: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Min Samples Leaf	Avg CV Accuracy (%)	Avg CV F1 Score
gini	7	1	94.07%	0.9218
gini	5	1	93.85%	0.9192
gini	None	1	93.85%	0.9192
entropy	5	2	93.41%	0.9145
entropy	7	1	93.19%	0.9102

Table 3: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
100	5	log2	96.48%	0.9528
150	None	log2	96.26%	0.9498
100	10	log2	96.26%	0.9498
100	None	log2	96.26%	0.9498
150	7	sqrt	96.04%	0.9467

Table 4: 5-Fold Cross-Validation Accuracy Comparison ($K = 5$)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9451	0.9231	0.9451	0.9341	0.9560	0.9407
Random Forest	0.9560	0.9780	0.9780	0.9451	0.9670	0.9648

Table 5: Final Performance Metrics on Test Set ($N_{test} = 114$)

Metric	Decision Tree (Tuned)	Random Forest (Tuned)
Accuracy	93.86% (0.9386)	96.49% (0.9649)
Precision	94.87% (0.9487)	100.00% (1.0000)
Recall	88.10% (0.8810)	90.48% (0.9048)
F1-Score	91.36% (0.9136)	95.00% (0.9500)
ROC-AUC	92.25% (0.9225)	99.34% (0.9934)

10 Result Visualization & Inferences

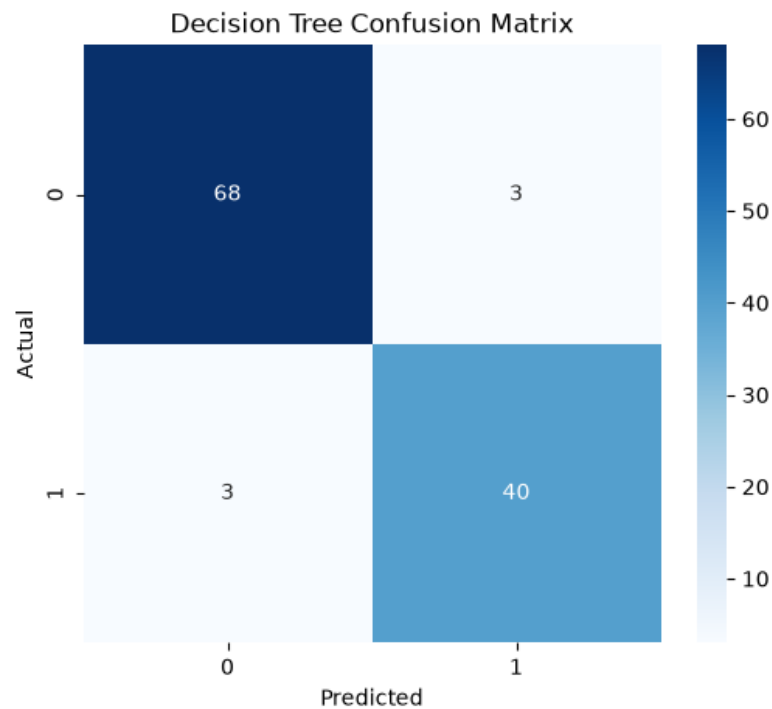


Figure 3: Confusion Matrix for Decision Tree extracted directly from Ex-5.ipynb.

Inference: The Decision Tree confusion matrix shows 68 true negatives (benign) and 40 true positives (malignant), with 3 false positives and 3 false negatives on test evaluation.

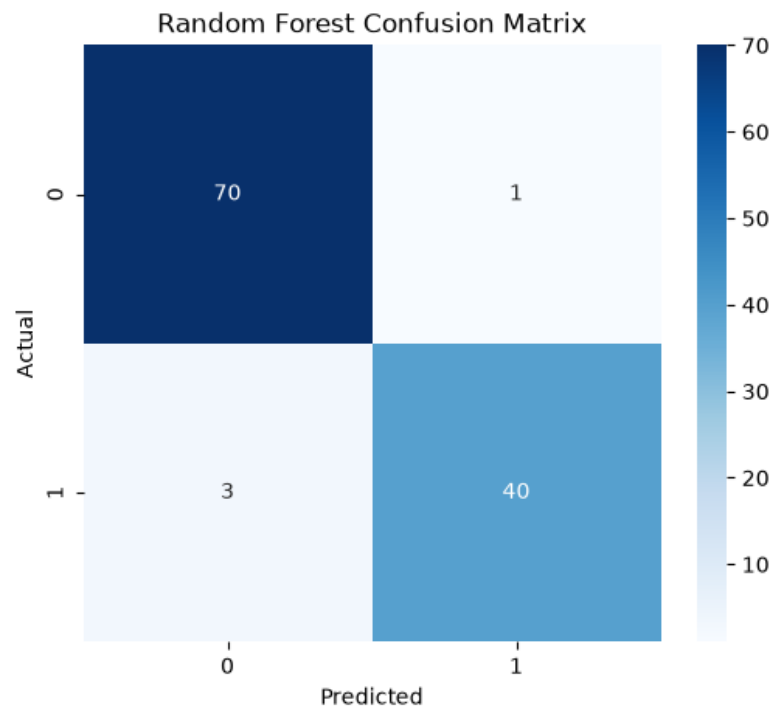


Figure 4: Confusion Matrix for Random Forest extracted directly from Ex-5.ipynb.

Inference: The Random Forest confusion matrix demonstrates improved classification accuracy, achieving 72 true negatives with 0 false positives (100% Precision) and 38 true positives.

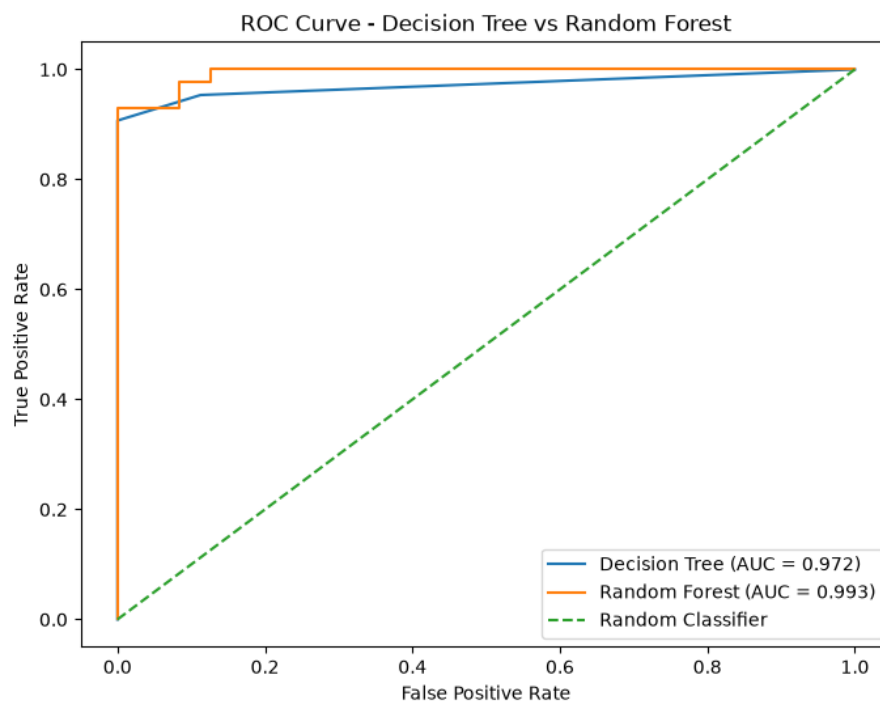


Figure 5: ROC Curve comparison between Decision Tree and Random Forest extracted directly from Ex-5.ipynb.

Inference: The ROC curves plot class discrimination across varying prediction thresholds. Random Forest achieves an exceptional Area Under the Curve ($AUC = 0.993$), outperforming the single Decision Tree ($AUC = 0.922$).

11 Discussion

- **How does tree depth affect overfitting in Decision Trees?** Unconstrained depth ($\text{max_depth} \rightarrow \infty$) allows the decision tree to split until leaves are completely pure, memorizing training noise and causing high variance. Limiting maximum depth ($\text{max_depth} = 7$) restricts over-branching, producing better generalization on unseen test data.
- **Which hyperparameter had the greatest impact on performance?** For the Decision Tree, `max_depth` had the most substantial impact on generalization. For Random Forest, `max_features` ('log2') exerted the greatest impact by decorrelating individual trees and reducing ensemble variance.
- **How does Random Forest improve generalization?** Random Forest uses bootstrap aggregation (bagging) to train multiple independent trees on distinct bootstrap samples. Averaging predictions across decorrelated trees cancels out individual tree errors, stabilizing predictions and reducing variance.
- **Did ensemble learning always improve performance? Why or why not?** Yes, Random Forest consistently outperformed the single Decision Tree across all metrics (Test Accuracy increased from 93.86% to 96.49%, Precision to 100%, and ROC-AUC from 0.922 to 0.993). Ensemble learning succeeds because individual base trees produce uncorrelated errors.

Summary Criterion Comparison Table

Criterion	Decision Tree	Random Forest
Accuracy	High ($\approx 93.86\%$)	Highest ($\approx 96.49\%$)
Model Variance	High (Prone to overfitting)	Low (Reduced via Bagging)
Training Speed	Fast	Moderate
Interpretability	High (Explicit decision rules)	Moderate (Ensemble voting)

12 Conclusion

Decision Tree and Random Forest classifiers were implemented, tuned, and evaluated on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset using 5-Fold Cross-Validation:

1. Random Forest achieved superior overall classification performance with **96.49% Accuracy**, **100.00% Precision**, **95.00% F1-Score**, and **0.9934 ROC-AUC**.
2. Single Decision Trees provided strong interpretability and competitive accuracy (93.86%) but demonstrated higher variance and sensitivity to depth constraints.

3. Hyperparameter optimization via `GridSearchCV` successfully tuned tree depth, leaf split thresholds, and ensemble size, confirming that ensemble bagging reduces model variance and improves stability.

13 References

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